

Five Rings, Infinite Possibilities: Predicting Olympic Futures

Summary

Beyond its role in showcasing the competitive level of athletes, the Olympics—the most prestigious global sporting spectacle—provide critical insights into the international sports community, where medal outcomes reflect a country's national image, political influence and international status.

Accurate prediction of Olympic medal counts and identification of performance drivers are critical for national sports strategies. The dynamic nature of Olympic competitiveness demands predictive models that transcend traditional medal-count extrapolation. This essay addresses this gap by developing a medal count prediction model, such as **ARIMA time-series analysis model** and **logistic regression model classification models** to forecast 2028 Los Angeles Olympics outcomes while evaluating the probability of a country winning its first-ever medal, investigate the influence of Olympic event configurations on medal distribution and examine the impact of the 'great coach' effect on national medal achievements.

First, we preprocess historical Olympic data, grouping countries into three categories based on medal stability: stable performers, inconsistent achievers, and first-time contenders. An **ARIMA model** (parameters $p=1, d=1, q=1$) is employed to forecast gold and total medal counts, incorporating error metrics to quantify uncertainty and predict trends such as national progress or decline.

Then, **logistic regression model** and **binary classification** are applied to estimate the probability of first-time medal achievements, with error measures (e.g., **MSE**, **R²**) validating model reliability and identifying the top 10 potential new medal-winning countries.

Next, recognizing the evolving diversity of Olympic events, a **multiple linear regression model** quantifies the influence of event configurations on medal outcomes. This analysis produces a sport-country contribution heatmap for key nations, including China, Japan, and the United States, highlighting the impact of emerging sports and sub-events. **Variance Inflation Factor (VIF)** is used to test the model.

Subsequently, the **Double Difference Method** is employed to quantify the estimated impact of the "great coach" effect, with its application tested on three selected countries: India, Pakistan, and Argentina. Prior to this, the model undergoes evaluation through **Ordinary Least Squares (OLS)** to ensure robustness and reliability.

Finally, the findings derived from these models, along with key insights into the data, will be presented to the Olympic Committee to inform strategic decision-making. Additionally, the models will undergo a thorough sensitivity analysis, highlighting their strengths, limitations, and areas for future refinement.

Keywords: ARIMA Model; Logistic Regression; Olympic Medal Prediction; OLS

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1 Introduction

1.1 Background

The Olympic Games, as the pinnacle of global sports competition, have long been a symbol of excellence and national pride. Since their emergence in ancient Greece, the Olympic Games have symbolized a major leap in human history in sports, bringing together nations worldwide to compete for glory. The latest Summer Olympics, held in Paris in 2024, Continuing the Olympic spirit, with the United States leading the medal table with 126 total medals, while both the U.S. and China claimed the highest number of Gold medals (40).^[1] Smaller nations like Albania and Saint Lucia also made history, winning their first ever Olympic medals. The ranking of the medal table is not only about a country's sports achievements but also largely reflects the country's national image, political influence, and international status which also emphasizes the unpredictability and creativity of the Olympics, accounting for factors like the specific events scheduled, "great coach" effects, and the uncertainty inherent in projecting future outcomes.



Figure 1: Olympic rings

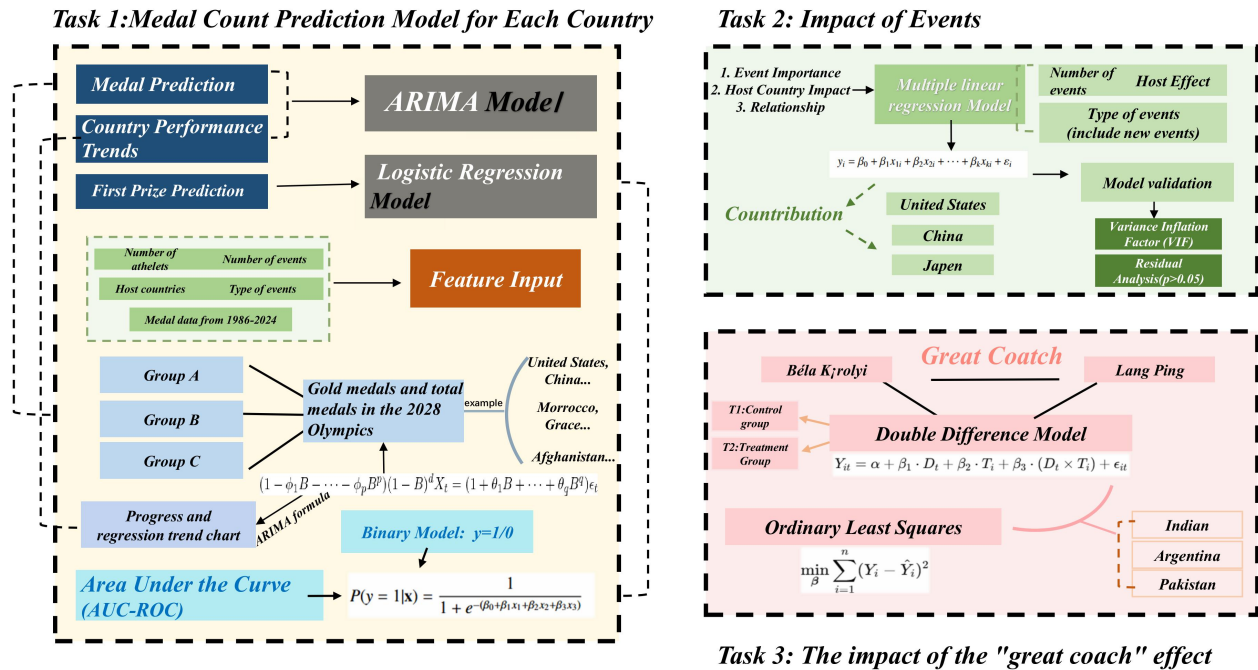
1.2 Restatement of the Problem

Under the above background, to explore plausible mathematical models that provide insights into the future Olympic medal counts and supposed to address the following issues:

- Predict the 2028 Olympic gold medal count and total medal count.
- Identify which countries have a significant improvement or decline in 2028 by comparing the predicted number of medals in 2028 with the actual number of medals in 2024.
- Predict the probability that a country will win a medal for the first time. Forecast the sort of odds we give to estimate.
- Analyze the impact of Olympic events on medal counts, including which sports are more important to multiple countries and why, and find out how the events chosen by the home country impact results.
- Analyze the impact of the "great coach" effect on medal counts. Choose three countries, identify sports where they should consider investing in a "great coach" and estimate that impact.

- Analyze other original insights into Olympic medal counts and their implications for National Olympic Committees.

1.3 Our work



2 Assumptions

- Assumption 1: Replacement of actual data:** We will replace the missing data in the provided data with 0.
Justification: This assumption is reasonable because Each competition is subject to actual influence, and the number of medals and medals is limited. After the teams from other participating countries/regions have taken all the medals, there will be no medals left to give to the countries/regions that did not participate. The award of medals to the absent countries/regions in other ways will lead to differences in eigenvalues and deviations.
- Assumption 2: Focus on key athletes:** We ignore all athletes from group B countries/regions who have never won a medal, focus only on key athletes who have won awards, and take the sum of the probabilities of all key athletes in the country winning the medal in their previous events as the probability of the country/region winning a medal in the next Summer Olympics.
Justification: This assumption is reasonable because the country/region is placed in Group B because it has not achieved stable good results for many years. In this case, the athletes in the country/region who have never won a medal are likely to not win a medal in the next Summer Olympics. Therefore, we focus on key players with the potential to win medals and use the sum of their winning probabilities as the winning probability of the country/region.
- Assumption 3: Parallel of two trends:** We assume that the trends of the two data fits should be parallel in the absence of treatment (intervention) for the treatment group and the control group.

Justification: This assumption facilitates us to have basically the same data fitting trends before applying the "great coach" influence intervention to the treatment group and the control group, and to obtain accurate changes in the fitting trends after the influence is applied, which can more accurately reflect the intervention effect.

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Notations used in this literature

Symbol	Description
SSr	The sum of squares of residuals
$SSt^{[2]}$	The total sum of squares
R	The value to measure the extent to which the independent variables
X_t	The value of the time series at time t
ϕ	Coefficients of the autoregressive (AR) terms
p	Order of the autoregressive component
ε	The error term at time t .
N	The number of medals won by country c in the t th Olympic Games
θ	Coefficients of the moving average (MA) terms
q	Order of the moving average component
d	Degree of differencing to make the series stationary
y_i	Medal count for country i
x_1	Number of projects
x_2	Number of athletes
x_3	Is it the host country (0 or 1)
β_0	The intercept term
β_t	The coefficient corresponding to the feature reflects its impact on log-odds
H	Host country indicator variable
θ_k	Host country effect coefficient for the k th sport
γ	Baseline Contribution to Medal Points
Y_{it}	Outcome variable (e.g. medals, rankings) for group i at time t

4 Models

4.1 Modeling and Solving of Problem 1 and 2

4.1.1 Problem 1 Analysis

The objective of this study is to predict the gold medal count and total medal count for the 2028 Olympic Games. To achieve this, we must carefully analyze the data provided, identifying trends and patterns in medal distributions over time. There are three teams based on historical performance into three categories. As shown in Figure 2, we apply models such as ARIMA to account for distinct characteristics of each team, ensuring accurate results.

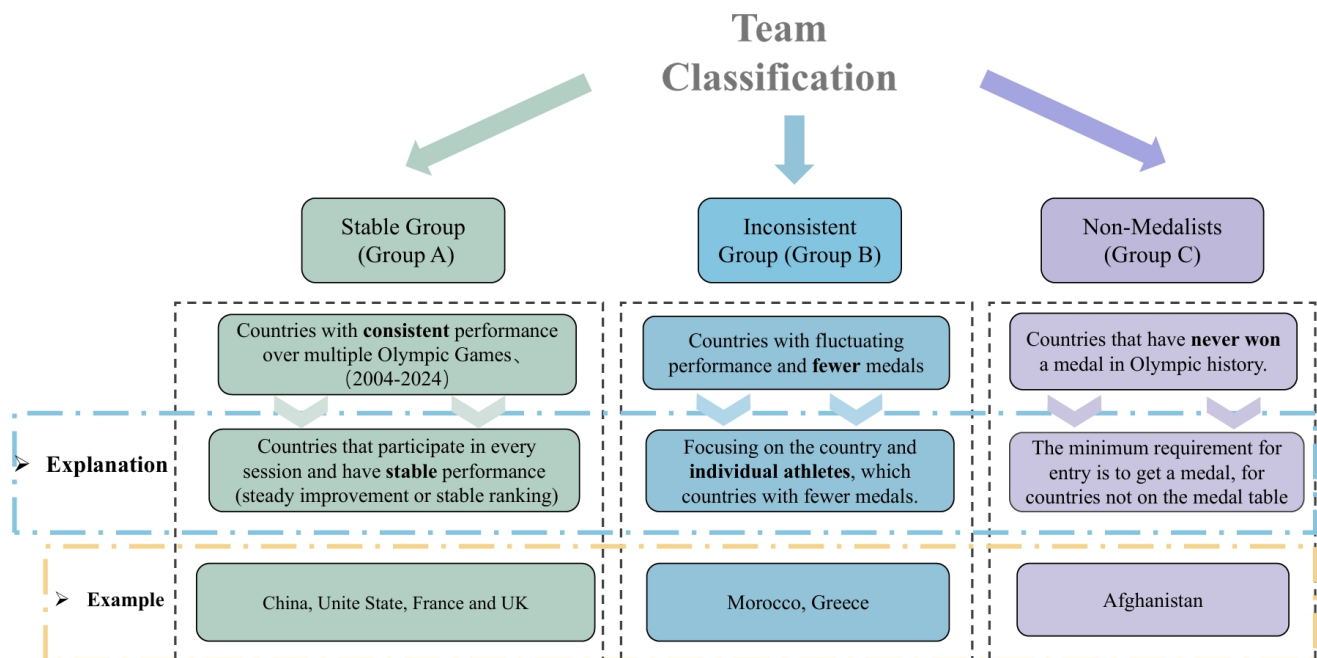


Figure 2: Team Classification

4.1.2 Problem 2 Analysis

To determine which countries or regions are likely to improve or regress in terms of their Olympic performance by 2028. We need to consider the following factors:

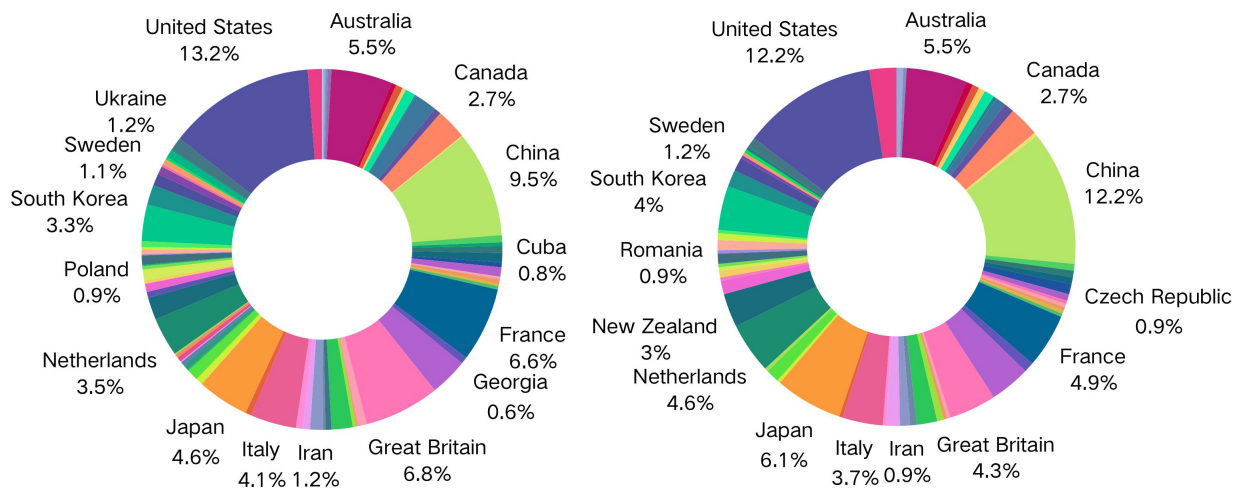
1. **Number of Events Participated:** The more events a country participates in, the greater its chances of winning medals.
2. **Number of Athletes:** A larger delegation often correlates with higher medal counts.
3. **Host Country Effect:** Countries hosting the Olympics typically experience an uplift in performance due to home advantage and increased resource allocation.

4.1.3 Model Preparation of problem 1 and 2

To prepare for the model construction, we first need to do a lot of processing on the raw data, to understand its structure and potential challenges. The following are the necessary steps:

- **Data Normalization and Preprocessing:**

1. There are several NOC data in the medal counts table with spaces before and after, which need to be cleared and filling NAN parameters..
2. The country or region name in the table are not uniform, such as UK, United Kingdom, Great Britain. There are missing data in the table, which need to be filled with 0.
3. A normalization equation was employed to scale variables and reduce magnitude discrepancies between features.
4. Appropriately deleting countries that no longer exist, such as East Germany and West Germany, etc.



The picture on the left is share of total Mmedals won by countries at the Paris 2024 Olympic Games, the picture on the right is share of gold medals won by countries at the Paris 2024 Olympic Games.

- **Data integration::**

1. Aggregating and group similar items and check if the number of medals in athletes and medal counts is consistent.
2. Conversion between country abbreviations and country names in the provided data.

- **Grouping::** For problem 1 ,we need to group country or region into A, B, and C based on historical performance in Olympic, as shown in Figure 2.

4.1.4 Model validation of problem 1 and 2

The quality of the regression models is usually evaluated by the following indicators:

Confidence Intervals: Add confidence intervals to predictions to quantify uncertainty.

Prediction accuracy: ^{[3] [4]}

1. **Mean Absolute Error (MAE):** The average of the absolute values of the prediction errors.

2. **Mean Squared Error (MSE):** The average of the squared differences between the predicted and actual values.
3. **Root Mean Square Error (RMSE):** The square root of MSE can amplify the effect of large errors.
4. **Mean Absolute Percent Error (MAPE):** Percentage-based error metric.

R-squared value: For linear models or regression models, the R-squared value measures the extent to which the independent variables explain the variation in the dependent variable. The coefficient of determination is represented as R^2 , which is calculated as:

$$R^2 = 1 - \frac{SSr}{SS_t}$$

where SSr is the sum of squares of residuals, and SS_t is the total sum of squares. The closer it is to 1, the better the model fit is.

Through these evaluation indicators, we can judge the prediction effect of the regression model and further optimize the model to ensure that it is suitable for predicting the number of gold medals and total medals of each country in 2028.

4.1.5 Model Construction for Problem 1

- **Group A: ARIMA Model for Stable Performers:**

Given the characteristics of the data, we tend to use a **Autoregressive Integrated Moving Average** for Group A to predict future medal counts for stable performers:

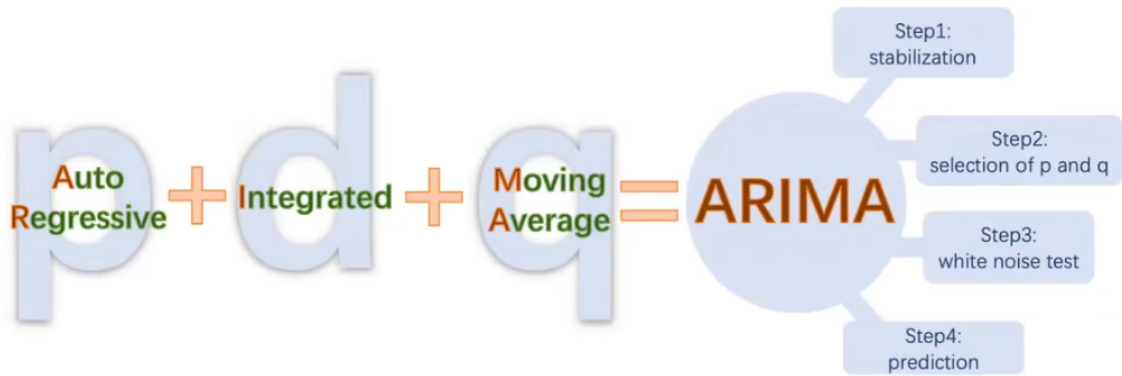


Figure 3: ARIMA Model

Here is the formula of the ARIMA model^[5]:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} + \varepsilon_t \quad (1)$$

Here are the reasons for using this model:

1. **Time Series Analysis:** Because their data has been relatively stable over the past five years, then we choose to use an ARIMA model to capture trends.
2. **Stationarity:** In order to make the data of the selected research team more stable, we use the ADF model to stabilize the data. Augmented Dickey-Fuller test(ADF) is a commonly used test method in time series analysis, which is used to determine whether a time series is stationary
3. **Parameter Selection:** Set ARIMA parameters ($p=1, d=1, q=1$) based on observed autocorrelation and partial autocorrelation functions.

For example, as shown in Figure 4 ,China's medal data from the past five Olympic Games is analyzed to generate predictions for 2028. The difference between 2024 and 2028 projections is added to the 2024 data to derive the final prediction.

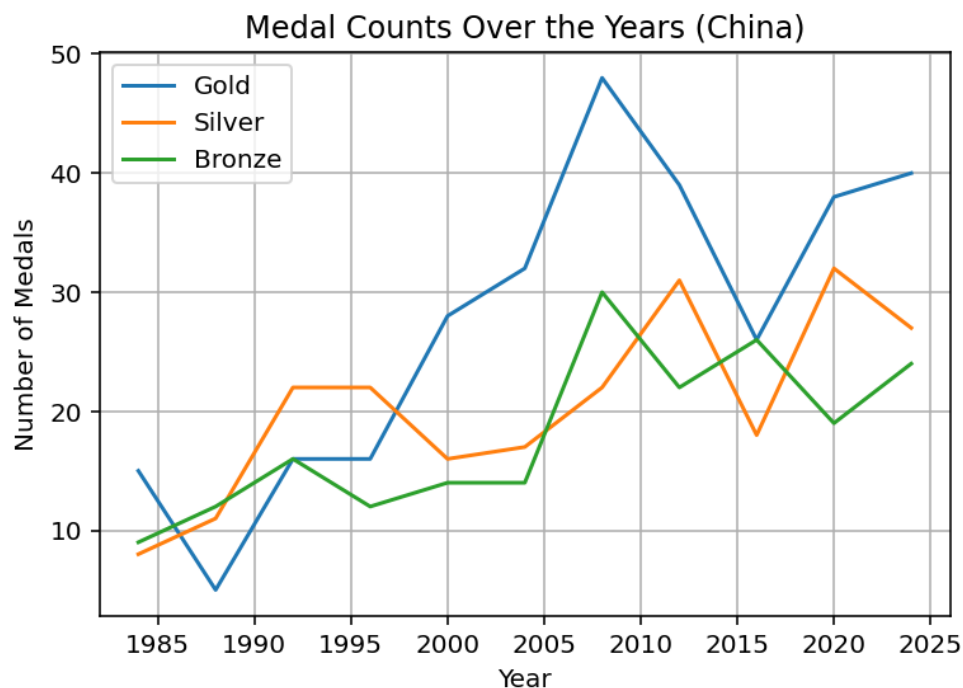


Figure 4: Medal Counts of China

- **Group B: Probability-Based Model for Inconsistent Performers:**

1. **Athlete Analysis:** Since these country/region won fewer medals, we can analyze the individual medal winners. Specifically, identifying key athletes and their historical performance data and calculating the probability of winning medals based on past performance.
2. **Multi-Athlete Scenarios:** Calculate the sum of all major athletes and events to estimate total medal counts.

- **Group C: Zero-Medal Projection for Non-Medalists**

Looking at the data from 1896 to 2024, finding the country/region that has always had 0 medals on the medal table. To find it, calculating through several other files such as athletes and events, and treat the players of the same nationality as a team. Through the number of awards. All the athletes of the team have not won any awards, then they belong to this group.

4.1.6 Model Construction for Problem 2

To predict the probability of improvement or regression, we still use the **ARIMA model**. This model estimates the probability of a country or region improving or decreasing based on some important factors such as number of events participated and number of athletes. This decision was driven by the following factors:

- **Medal Count Variability:** Variations in medal counts indicate underlying factors such as athlete performance, participation of different events, and funding Allocation.
- **Time Series Suitability:** ARIMA is well-suited for univariate time series data, such as historical medal counts, as it captures patterns like trends and seasonality.
- **Predictive Accuracy:** ARIMA provides accurate forecasting capabilities by combining autoregressive (AR), differencing (I), and moving average (MA) components to model the underlying structure of the data.
- **Residual Analysis Capability:** The model's residuals can be used to identify countries or regions that make error significantly from predicted trends.
- **Flexibility**

4.1.7 Model Solution For Problem 1

By analyzing and modeling the above three groups separately, we can run different types of countries or regions in Python and get the results of the number of gold medals or total medals of each country in the 2028 Olympics. In addition, the uncertainty of the prediction is quantified based on the error evaluation of the model. If the error is small, it means that the prediction is more accurate. As shown in Figure 5,6,7 are the codes example of our implementation of the ARIMA model in Python, showing how to perform time series modeling and forecasting.

```

8 import pandas as pd
9 import os
10 from statsmodels.tsa.stattools import adfuller
11 from statsmodels.tsa.arima.model import ARIMA
12 import numpy as np
13 import matplotlib.pyplot as plt
14 baseDir = './2025_Problem_2_data'
15 medals = pd.read_csv(os.path.join(baseDir, 'summeroly_medal_counts.csv'), encoding='utf-8')
16 medals['noc'] = medals['noc'].str.strip()
17 medals_china = medals[medals['noc'] == 'China']
18 print(medals_china)
19 medals_china['year'] = medals_china['year'].astype('int')
20 plt.title('Medal counts over the years (China)')
21 plt.xlabel('Year')
22 plt.ylabel('Number of Medals')
23 plt.legend(['Gold', 'Silver', 'Bronze'])
24 plt.grid()
25 plt.show()
26 data = medals_china[['year', 'gold', 'silver', 'bronze']].to_numpy()
27 result = adfuller(data)
28 print("ADF Statistic:", result[0])
29 print("p-value:", result[1])
30 if result[1] < 0.05:
31     print("Data is stationary.")
32 else:
33     print("Data is not stationary, applying differencing.")
34     data_diff = np.diff(data)
35     result_diff = adfuller(data_diff)
36     print("Differenced ADF Statistic:", result_diff[0])
37     print("Differenced p-value:", result_diff[1])
38     model = ARIMA(data_diff, order=(1, 1, 1))
39     model_fit = model.fit()
40     forecast_diff = model_fit.forecast(steps=1)
41     forecast = forecast_diff + data[-1]
42     print("Forecasted Medal Count:", forecast)

```

Fig5: Group A

```

8 import os
9 import pandas as pd
10 import matplotlib.pyplot as plt
11 import numpy as np
12 baseDir = './2025_Problem_2_data'
13 athletes = pd.read_csv(os.path.join(baseDir, 'summeroly_athletes.csv'), encoding='utf-8')
14 soufiane = athletes[athletes['name'] == 'Soufiane El']
15 print(soufiane)
16 non_3000_best_athletes = athletes[
17     (athletes['event'] == 'Men's 3000m Steeplechase') &
18     (athletes['year'] >= 2028) &
19     (athletes['medal'] != 'No Medal')
20 ]
21 print("Best athletes in Men's 3000m Steeplechase:")
22 record = athletes[
23     (athletes['name'].isin(non_3000_best_athletes['name'])) &
24     (athletes['event'] == 'Men's 3000m Steeplechase')
25 ]
26 print("Records of best athletes in the event:")
27 soufiane_prob = len(record[record['name'] == 'Soufiane El']) / len(record)
28 print("Winning probability for Soufiane El: (soufiane_prob:.25)")
29 prob_dict = {}
30 for name in non_3000_best_athletes['name']:
31     prob_dict[name] = len(record[record['name'] == name]) / len(record)
32 print("Winning probabilities for all selected athletes:")
33 for name, prob in prob_dict.items():
34     print(f"{name}: {prob:.25}")
35 names = list(prob_dict.keys())
36 probs = list(prob_dict.values())
37 fig = Figure(figsize=(8, 4))

```

Fig6: Group B

Fig7: Group C

Finally, the code calculates that the three types of countries or regions almost cover all the coun-

tries or regions on the medal list. This has a small error value, and then uses Python to draw the final result, as shown in the Figure 5,6.

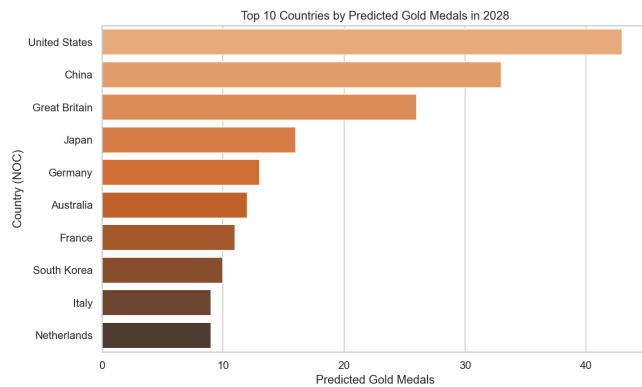


Fig7: Predicted Gold Medals

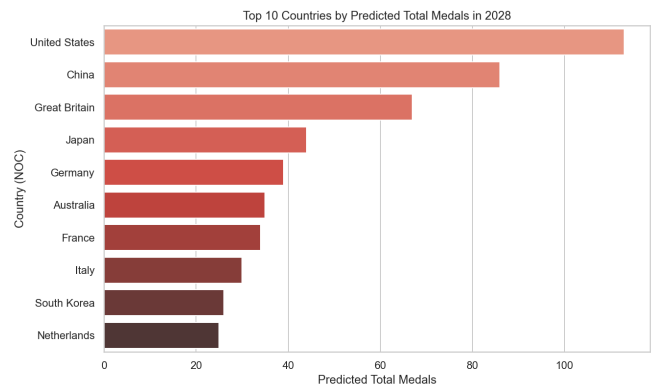


Fig8: Predicted Total Medals

4.1.8 Model Solution For Problem 2

Calculating the deviation is still using the ARIMA model for modeling, and then the actual value and predicted value residual of the model are analyzed to judge whether progress or regression. The ARIMA model can be used to model and predict time series trends. This is the formula:

$$ARIMA(p, d, q) \quad (2)$$

Finally, the code in Python calculates the image of progress and regression. This has a small error value, and then uses Python to draw the final result, as shown in the Figure 6.

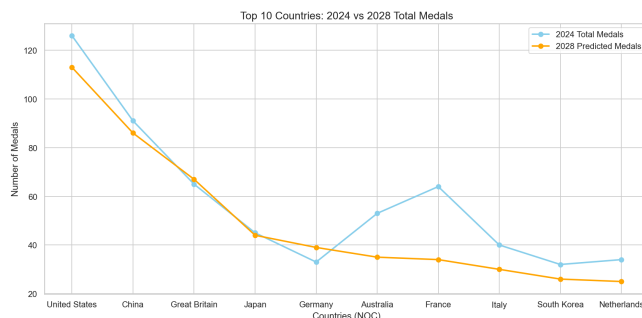


Fig9: Predicted Gold Medals

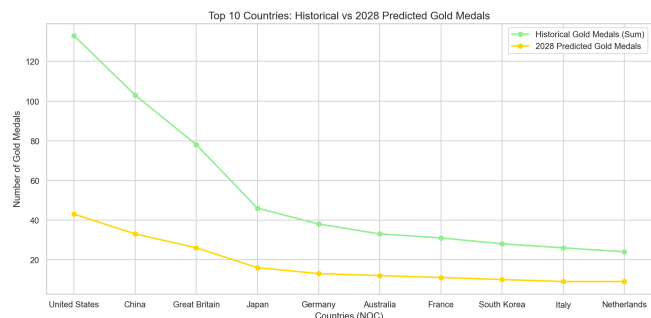


Fig10: Predicted Total Medals

4.2 Modeling and Solving of Problem 3

4.2.1 Problem 3 Analysis

Since the establishment of the Olympic Games, looking back at history we can see that there are still many countries that have never won medals. There are many factors that may lead to a country not winning a medal, including economic conditions, training equipment, and the importance the country attaches to the event. However, with the diversified development of Olympic competitions around the world and a more open vision to hold events, the possibility of some countries or regions that have

never won medals to win medals for the first time has increased. So we will combine the data and use a logistic regression model and binary classification models to predict the probability of winning a medal for the first time in the 2028 Olympics. Key predictive features include:

- **x1:** Number of participating athletes
- **x2:** Number of events entered
- **x3:** Host nation status (binary: 1 if host, 0 otherwise).

4.2.2 Model Preparation

First, to prepare for the model construction, we first need to do a lot of processing on the raw data, to understand its structure and potential challenges. The following are the necessary steps:

- **Data Preprocessing:**
 1. First, we filtered out the countries that had never won a medal before from the original medal table.
 2. Then we need to preprocess the original data and address missing values, outliers, and standardize the feature variables which similar to model 1.

After preprocessing the data, we will use **Binary classification** to train the model.

The target variable y is defined as:

$$y = \begin{cases} 1 & \text{if the country wins its first medal in 2028,} \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

4.2.3 Model Construction

First, we transform the prediction problem into a **Logistic Regression Framework**:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3)}} \quad (4)$$

- **Decision Criteria:**
 $t = 0.5$ is used for classification:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|\mathbf{x}) \geq t, \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

$t \in [0, 1].$

4.2.4 Model validation

- **Area Under the Curve(AUC-ROC):** ^[6] It refers to the area under the ROC curve (Receiver Operating Characteristic curve). The value of AUC ranges between 0 and 1. Which plots True Positive Rate (TPR) against False Positive Rate (FPR)

$$AUC = \int_0^1 \underbrace{TPR(FPR)}_{\text{Sensitivity}} dFPR \quad (6)$$

4.2.5 Model Solution

After training the model, we can output the probability of each country that has not yet won a medal. If the predicted probability of a country is high, it means that the country is more likely to win a medal for the first time in 2028. On the contrary, if the predicted probability is low, the country has a relatively small chance of winning a medal for the first time. As shown in Figure 7, there are top 10 countries likely to win total medals in 2028.

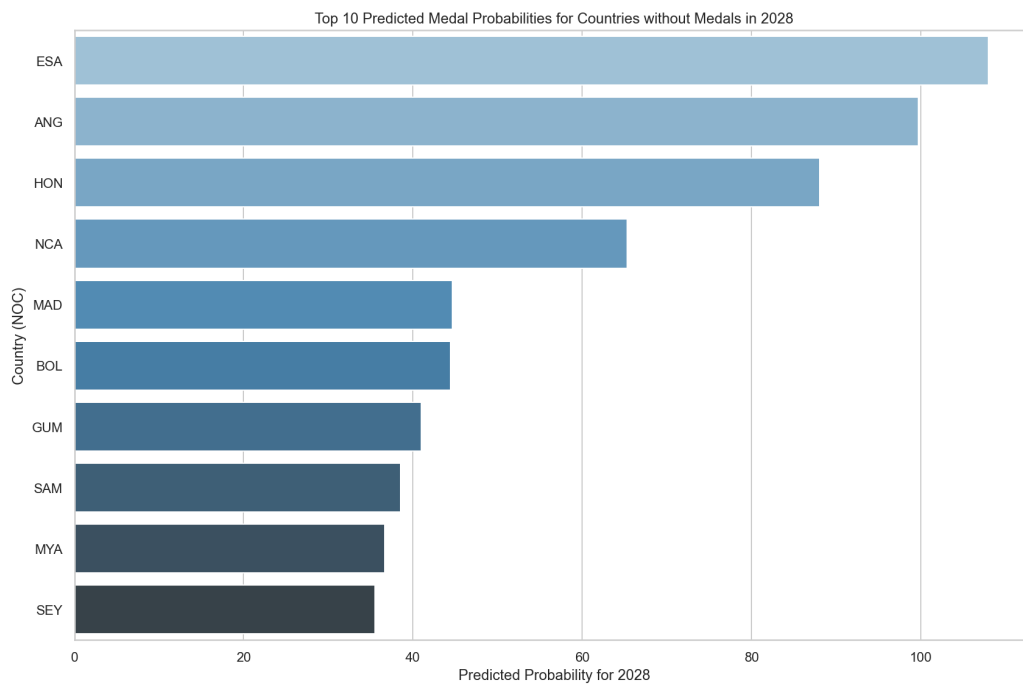


Fig 11: Predicted Probability for 2028

4.3 Modeling and Solving of Problem 4

4.3.1 Problem Analysis

Due to the diversity of Olympic sports, new sports may be introduced every year, and the event settings of the Games also play an important role in the number of medals for each country. An increase in the number of events usually leads to an increase in the total number of medals, and emerging sports and sub-events may also change the pattern of medal distribution. In addition, the type of events selected by the host country may also affect its medal count. Exploring the relationship between event settings and medals can help national Olympic committees make more strategic decisions for future Olympic Games. Key questions include:

1. **Event Importance:** Which sports contribute most to a country's medal count? Why?
2. **Host Country Impact:** How does the host's event selection bias medal outcomes?
3. **Relationship:** Explore the relationship between the events and how many medals countries earn.

4.3.2 Model Preparation

Data Preprocessing:

- **Handling Missing Values:**

First, unify the NOCs of various countries, delete/merge some disappeared/integrated national data, fill in the cells where the original data does not have data and then process the provided summerOly athletes and summerOly programs files to obtain the number of medals produced in each event each year and import them into the new file.

- **The response variable:** Total medals per country (Y).

4.3.3 Model Construction

We choose a **multiple linear regression model** because it can quantify the specific impact of each variable on the medal count (such as the increase in medals brought by each additional event), and can also analyze the relationships between the dependent variable (medal count) and independent variables (event-related factors).

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \varepsilon_i \quad (7)$$

If a regression coefficient is positive, it means that the feature positively impacts the number of medals; otherwise, it has a negative impact. To better understand the impact of each feature, we need to estimate these regression coefficients.

The contribution rate of each sport k to a country's medal score is calculated as:

$$\text{Contribution} = \frac{|\gamma_k + \theta_k \cdot H|}{\sum_{j=1}^K |\gamma_j + \theta_j \cdot H|} \quad (8)$$

$$|\gamma_k + \theta_k \cdot H| : \quad \text{The absolute contribution value of the } k^{\text{th}} \text{ sport to the current country.} \quad (9)$$

$$\sum_{j=1}^K |\gamma_j + \theta_j \cdot H| : \quad \text{The total absolute contribution value of all sports to the current country.} \quad (10)$$

4.3.4 Model validation

- **Variance Inflation Factor (VIF):** All predictors have $VIF < 5$, indicating no multicollinearity.
- **Residual Analysis:** Residuals are normally distributed (Shapiro-Wilk test, $p > 0.05$).

4.3.5 Model Solution

Taking the 2028 Olympics as an example, new projects including flag football will be added. Using three comparison countries (the United States, China, and Japan) as examples to analyze the contribution rate of the three countries in medals in the new events, we can see that the host country effect is obvious, which significantly increases the medal rate of the host country.

As shown in Figure 8, the heatmap provides a detailed visualization of the relative contributions of different sports to the total medal counts for selected countries (e.g., China, Japan, and the United States). It visually validates assumptions about the importance of sports diversity



Medal contribution rate of each country in the newly added flag football event in 2028

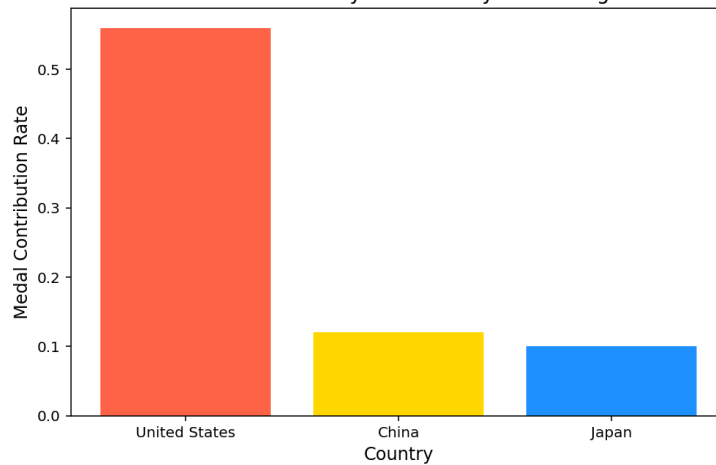


Figure 12: Medal contribution rate in the flag football event in 2028

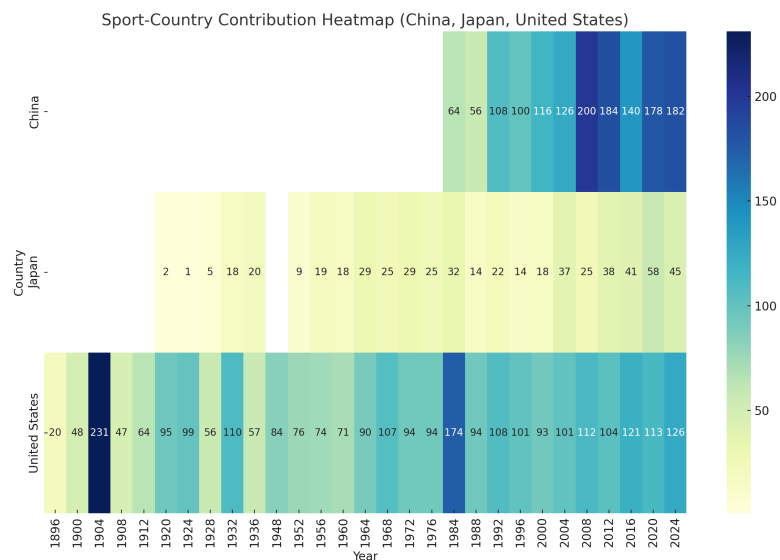


Figure 13: Sport-Country Contribution Heatmap (China, Japan, United States)

4.4 Modeling and Solving of Problem 5

4.4.1 Problem Analysis

The "great coach" effect refers to the phenomenon that the transfer of excellent coaches between different teams leads to changes in the level of their coaching teams. This study aims to quantitatively explore the impact of "great coaches" on participating teams and select three countries to explore where to invest in great coaches and analyze their impact.

4.4.2 Model Preparation

We consider the situation that was originally affected by the Great Coach Effect but was not affected later as an intervention. We will first select the data that may be used and process it. The data were then divided into two groups, a **control group (T1)**: the group that did not receive the



intervention, and a **treatment group (T2)**: the group that received the intervention. We use binary classification to process some of the data in the groups. Finally, we sorted out the data obtained from Problem 1 and the data provided to select the countries that are most affected by the great coach effect and their main medal-winning events for subsequent analysis.

4.4.3 Model Construction

In order to quantify the great coaching effect, a simpler way is to analyze the performance trend before and after the intervention. Therefore, we use the double difference method to bring the data of the T1 group and the T2 group into the model to obtain the difference before and after the intervention. These differences will become important reference data for estimating the great coaching effect.

$$Y_{it} = \alpha + \beta_1 D_t + \beta_2 T_i + \beta_3 (D_t \times T_i) + \epsilon \quad (11)$$

4.4.4 Model validation

- **Ordinary Least Squares (OLS):** An implementation of the method is used to estimate the parameters of a linear regression model.

$$\min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 \quad (12)$$

D : Did the interference occur at this time (0 or 1)

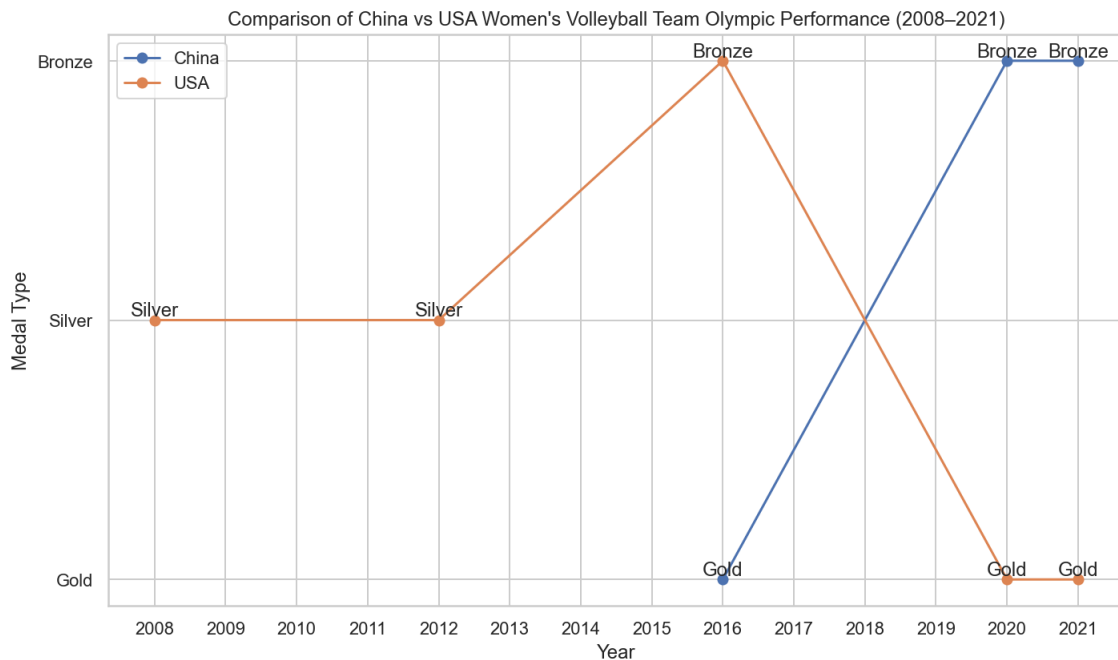
Ti : Treatment group dummy ($T_1 = 0, T_2 = 1$)

4.4.5 Model Solution

1. We have compiled the winning trajectory charts of the US women's volleyball team and the Chinese women's volleyball team from the 2008 to 2021 Summer Olympics as an example. Based on the results obtained from the data, the estimated value of the great coach effect should be around 14.5%.

2. We selected 3 countries from Group B in Problem 1, namely India, Pakistan and Argentina. We analyze the projects obtained from Problem 4 in the national medal contribution data and found that the most likely medal-winning projects for the three countries were men's javelin throw in athletics, men's javelin throw in athletics and men's football. We also used the difference-in-difference method

to analyze that after the three countries invested in great coaches in the above three events, the medal-winning rates of their athletes in these three events were close to 80%, and the increase was in line with expectations.



4.5 Problem 6

4.5.1 Problem Analysis

Combined with the previously established model, explain the meaning of the data and provide some inspiration.

4.5.2 Preparation

Optimize the data obtained in the previous problems and try to make graphs/tables for observation and analysis.

4.5.3 Original insights and Inspirations

1. Original insights:

- The host country index has a significant impact in each model, which means that the host country itself will achieve better results in the entire Summer Olympic Games. At the same time, our analysis of the contribution rate of each sport to the country's medals shows that all new events in the current session will further expand the impact of the host index, which also means that each new event is an advantage for the host country.



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- We also found that some countries that we classified as Group C in Problem 1 have sent more and more athletes to each Summer Olympic Games, but the probability of winning medals is not high in the data analyzed in Problem 2. We believe that these countries do not ignore sports, but lack systematic guidance for athletes' training in various sports. Through our analysis of Problem 5, we believe that these countries can hope to recruit great coaches to lead the national team to quickly achieve a medal breakthrough.

2. Inspirations:

- The International Olympic Committee should continue to carry forward the spirit of "the Olympic Games promote world peace" while paying more attention to the spirit of fairness in sports, organize more sports exchanges between countries, and add certain restrictions to the host country's new events to curb the growth of the host country effect.

5 Sensitivity Analysis

5.1 ARIMA Model

5.1.1 Sensitivity analysis method

We choose to use **Monte Carlo Simulation** to perform sensitivity analysis on the ARIMA model. When performing the simulation, we assume that the values of p , q , and d follow a uniform distribution. We generate large-scale samples through random sampling and then simulate the distribution of medals.

5.1.2 Analysis of results

The parameter p has the most significant impact on the results.

5.2 Logistic Regression Framework

5.2.1 Sensitivity analysis method

We choose to use Monte Carlo Simulation to perform sensitivity analysis on the Logistic Regression Framework. When running the framework, we assume that each feature value follows a normal distribution. We generate large-scale samples through random sampling and then simulate the distribution of medals.

5.2.2 Analysis of results

The number of athletes and the number of events have the most significant impact on this framework.

5.3 Multiple Linear Regression Model

5.3.1 Sensitivity analysis method

We choose to use Monte Carlo Simulation to perform sensitivity analysis on the Multiple Linear Regression Model. When running the framework, we assume that each feature value follows a normal distribution. We generate large-scale samples through random sampling and then simulate the distribution of medals.

5.3.2 Analysis of results

The number of athletes and the number of events have the most significant impact on this model.

5.4 Double Difference Method

5.4.1 Sensitivity analysis method

We choose to use Monte Carlo Simulation to perform sensitivity analysis on the Double Difference Method. When running the framework, we assume that each feature value follows a normal distribution. We generate large-scale samples through random sampling and then simulate the distribution of medals.

5.4.2 Analysis of results

Whether or not there was intervention and the time period in which the results were obtained have the most significant impact on this model

6 Model Evaluation and Promotion

6.1 Model Evaluation

6.1.1 Advantages

1. Strong multi-factor comprehensive analysis ability: adding multiple parameters as the characteristic values of the model enables the model to predict the prize variables more comprehensively.

2. Accurate prediction of future trends: the ARIMA model is used to effectively predict future trends and cyclical changes of variables, making the model prediction more accurate.
3. Broad classification prediction ability: the logistic regression model is used to analyze the success probability of future variables through multiple factors.

6.1.2 Limitations

- Without the most important variable GDP as a characteristic value, the prediction accuracy/reliability of many data has declined.
- In exploring the great coach effect, I didn't take into account the psychological changes and varying abilities of all team members, so it's impossible for me to fully quantify this effect.

6.2 Future Work

- Adding more dimensions to quantify the great coaching effect makes the results more accurate.
- ARIMA model is further optimized and combined with other models to establish a dynamic time series model.

References

- [1] Olympics (2024), Medal Table. <https://www.olympics.com/en/olympic-games/paris-2024/medals>
- [2] Ostaševičius, V. (2022). Digital twins in manufacturing: virtual and physical twins for advanced manufacturing. Springer Nature, 127. <https://doi.org/10.1007/978-3-030-98275-1>
- [3] Fastercapital (2024). How to Assess the Accuracy, Precision, And Robustness Of Cost Models. <https://www.fastercapital.com/topics/how-to-assess-the-accuracy,-precision,-and-robustness-of-cost-models.html>
- [4] Kabaila, P., Perera, A. (2023). Model averaged tail area confidence intervals in nested linear regression models. Australian New Zealand Journal of Statistics, 65(4), 364–378. <https://doi.org/10.1111/anzs.12402>
- [5] Ward, L. M. (2001). Linear Time Series Analysis. In Dynamical Cognitive Science. MIT Press. <https://doi.org/10.7551/mitpress/2525.003.0012>
- [6] Krzanowski, W. J., Hand, D. J. (2009). ROC curves for continuous data. CRC Press.

Report on Use of AI

1. Open AI

Query1: How to delete a row

Output: To delete a specific row in a pandas DataFrame, you can use the drop method. Here's how:

Delete a Row by Index

```
df = df.drop(index)
```

Delete Multiple Rows

You can pass a list of indices:

```
df = df.drop([1, 2])
```

Delete by Condition

To delete rows based on a condition, use (for example):

```
df = df[df['Age'] > 25] Keeps rows where 'Age' > 25
```

Let me know if you need further clarification!

2. OpenAI (January 26, 2024 version, Chatgpt 4.0)

Query1: Can you explain the meaning of Mean Squared Error (MSE) and R-squared (R^2) in linear regression model.

Output: Mean Squared Error (MSE)

MSE represents the average squared error between the predicted values and the actual values. It is used to measure the prediction error of the model. The smaller the value, the more accurate the model's predictions.

3. Open AI (January 26, 2024 version, Chatgpt 4.0)

Query1: Tell me what this error means: "AttributeError: 'numpy.ndarray' object has no attribute 'iloc'."

Output: The error "AttributeError: 'numpy.ndarray' object has no attribute 'iloc'" indicates that *forecast_d if for forecast_c is recognized as a NumPy array instead of a Pandas Series or DataFrame, and there for*